Introduction to Biophysics
Technische Universität München
Wintersemester 2025/2026
Prof. Julia Herzen, Prof. Andreas Bausch, Prof. Ulrich Gerland

Solutions to Exercise Sheet 6

Problem 6.1

You should advise the runner to breathe rapidly just before the race. Since a sprint will cause a lowering of blood and cell pH, the object of the pre-race routine would be to raise the pH with the idea that the runner could then sprint longer before feeling fatigue. Holding your breath or breathing rapidly both temporarily affect the amount of dissolved CO_2 in the bloodstream. Holding your breath will increase the amount of CO_2 and push the equilibrium to the right, leading to an increase in $[H^+]$ and a lower pH. By contrast, breathing rapidly will reduce the concentration of CO_2 and pull the equilibrium to the left, leading to a decrease in $[H^+]$ and a higher pH.

Problem 6.2

It seems counterintuitive that polymerization of free tubulin subunits into highly ordered microtubules should occur with an overall increase in entropy (decrease in order). But it is counterintuitive only if one considers the subunits in isolation. Remember that thermodynamics refers to the whole system, which includes the water molecules. The increase in entropy is due largely to the effects of polymerization on water molecules. The surfaces of the tubulin subunits that bind together to form microtubules are fairly hydrophobic, and constrain (order) the water molecules in their immediate vicinity. Upon polymerization, these constrained water molecules are freed up to interact with other water molecules. Their new found disorder much exceeds the increased order of the protein subunits, and thus the net increase in entropy (disorder) favors polymerization.

Problem 6.3

The reverse of the forward reaction is simply not a possibility under physiological conditions. The flow of material through a pathway requires that the ΔG values for every step must be negative. Thus, for a flow from liver glycogen to serum glucose the step from glucose 6-phosphate to glucose must have a negative ΔG. To simply reverse the forward reaction (that is, G6P + ADP $\rightarrow$ glucose + ATP, $\Delta G^\circ = 4.0$ kcal/mole) would require that the concentration ratio [glucose][ATP]/ [G6P][ADP] be less than 0.0015 in order to bring the reaction to equilibrium ($\Delta G = 0$).

$$
\begin{aligned}
\Delta G^{\circ} &= -RT\ lnK_{eq}\ (for\ a\ system\ at\ chemical\ equilibrium)\\
4.0\ kcal/mole &= -0.616\ kcal/mole\ ln\frac{[glucose][ATP]}{[G6P][ADP]}\ (at\ 37^{\circ}C)\\
ln\frac{[glucose][ATP]}{[G6P][ADP]} &= \frac{-4.0\ kcal/mole}{0.616\ kcal/mole} = -6.49\\
\frac{[glucose][ATP]}{[G6P][ADP]} &= 0.0015
\end{aligned}
$$

Inside a functioning cell, such as a liver cell exporting glucose, the concentration of ATP always exceeds that of ADP, but just for illustrative purposes let us assume the ratio is 1. Under these conditions the ratio [glucose]/[G6P] must be 0.0015; that is, the concentration of G6P must be nearly 700 times that of glucose. Since the circulating concentration of glucose is maintained at between 4 and 5 mM, this corresponds to about 3 M G6P, an impossible concentration given that the total concentration of cellular phosphate is less than about 25 mM.